Single Valid PIN Test Report

Project: Smart Home Alarm Controller — UVM Verification

Test: pin_valid_test

Purpose

Verify that the DUT accepts one valid keypad PIN and asserts pin_ok.

PIN Used

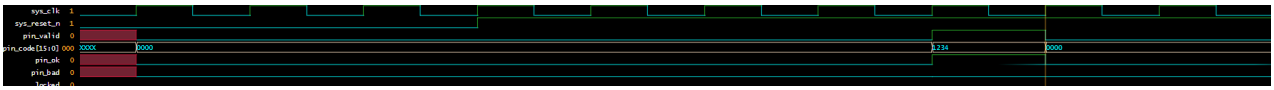
The sequence uses DEF_PIN from alarm_pkg. The project package defines the default PIN as 32'h1234. The sequence assigns req.pin_code = DEF_PIN, and the DUT compares the entered PIN with its configured PIN.

Result

PASS — The UVM checker reports Actual DUT pin_ok=1 at 75 ns and confirms that the DUT accepted the valid PIN. The keypad monitor observes pin_code=1234 at 85 ns.

Waveform Evidence

The waveform shows the valid PIN attempt and the DUT pin_ok pulse.



Complete UVM Log

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UVM_INFO alarm_output_passive_monitor.sv(71) @ 5: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 7: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 7: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 15: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 21: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 21: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 25: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 35: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 35: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 35: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 45: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 49: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 49: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO keypad_checker.sv(144) @ 50: uvm_test_top.alarm_env.keypad_chk [KEYPAD_VALID] Waiting for actual DUT pin_ok response
UVM_INFO pin_valid_sequence.sv(15) @ 50: uvm_test_top.alarm_env.keypad_agent.keypad_sequencer@@pin_seq [PIN_VALID_SEQ] START: Single valid PIN test
UVM_INFO alarm_output_passive_monitor.sv(71) @ 55: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 63: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 63: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 65: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 75: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_checker.sv(170) @ 75: uvm_test_top.alarm_env.keypad_chk [KEYPAD_VALID] Actual DUT pin_ok=1
UVM_INFO keypad_checker.sv(174) @ 75: uvm_test_top.alarm_env.keypad_chk [KEYPAD_VALID] PASS: DUT accepted the valid PIN
UVM_INFO sensor_monitor.sv(42) @ 77: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 85: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_monitor.sv(48) @ 85: uvm_test_top.alarm_env.keypad_agent.keypad_monitor [KEYPAD_MON] Observed pin_valid=1 pin_code=1234
UVM_INFO keypad_checker.sv(81) @ 85: uvm_test_top.alarm_env.keypad_chk [KEYPAD_VALID] Valid PIN stimulus observed: PIN=1234
UVM_INFO keypad_checker.sv(89) @ 85: uvm_test_top.alarm_env.keypad_chk [KEYPAD_VALID] Expected PIN_OK=0 | PIN=1234
UVM_INFO alarm_reference_model.sv(290) @ 85: uvm_test_top.alarm_env.ref_model [REF_MODEL] KEYPAD: VALID PIN expected, fail_count=0, locked=0
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UVM_INFO pin_valid_sequence.sv(34) @ 85: uvm_test_top.alarm_env.keypad_sequencer@@pin_seq [PIN_VALID_SEQ] Valid PIN attempt: PIN=1234
UVM_INFO sensor_monitor.sv(42) @ 91: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 91: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 95: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 105: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 105: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 105: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO pin_valid_sequence.sv(46) @ 105: uvm_test_top.alarm_env.keypad_sequencer@@pin_seq [PIN_VALID_SEQ] END: Single valid PIN stimulus completed
UVM_INFO pin_valid_test.sv(40) @ 105: uvm_test_top [PIN_VALID_TEST] Completed valid PIN test
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_objection.svh(1276) @ 105: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_report_server.svh(904) @ 105: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

UVM Keypad Verification Report
Invalid PIN Test

Smart Home Alarm Controller — Verification using UVM

1. Objective

Verify that the DUT rejects invalid PIN codes by asserting the actual pin_bad signal. This test focuses on invalid PIN rejection; lockout behavior is checked separately by pin_lockout_test.

2. Test stimulus

The pin_invalid_sequence applies three invalid PIN attempts, returning pin_valid low between attempts so each attempt is a separate rising-edge event.

Attempt	PIN code	Purpose
1	EDCB	Invalid PIN
2	A5A5	Invalid PIN
3	5A5A	Invalid PIN

3. Waveform

The waveform shows pin_valid and pin_code changing for each attempt. The actual DUT pin_bad signal produces a short HIGH pulse for each invalid PIN. The locked signal remains LOW in this test.

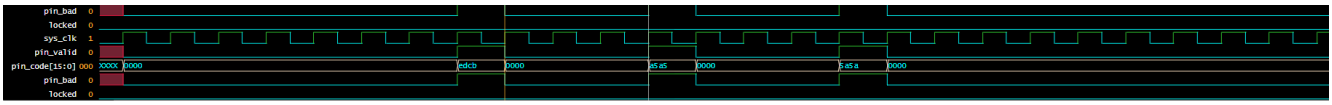


Figure 1. Waveform captured for the invalid PIN test.

4. Simulation log evidence

The checker observed all three actual DUT pin_bad responses:

- 75 ns — PASS: DUT rejected invalid PIN attempt 1/3
- 115 ns — PASS: DUT rejected invalid PIN attempt 2/3
- 155 ns — PASS: DUT rejected invalid PIN attempt 3/3

The keypad monitor later reports the corresponding stimulus values at 85 ns (EDCB), 125 ns (A5A5), and 165 ns (5A5A). The monitor's transaction observation time differs from the earlier combinational pin_bad pulse time.

5. Result and interpretation

PASS for invalid PIN rejection: all three invalid attempts caused the DUT pin_bad signal to assert. The waveform shows locked remaining LOW. This is not a lockout pass; lockout is outside this test's scope and is checked in pin_lockout_test.

The reference-model log expects lockout after the third failure, but the actual locked waveform signal does not assert in this run. The supplied pin_auth.sv contains an injected failure-counter bug (#5), which resets fail_cnt on pin_bad and therefore prevents the count from accumulating to the lockout threshold. Keep the RTL unchanged for bug-detection verification; the separate pin_lockout_test should expose this failure.

6. Conclusion

The invalid PIN test demonstrates that the DUT rejects each supplied incorrect PIN. It does not verify lockout activation. The continuously LOW locked signal is a result to investigate in the separate lockout test, not a failure of the invalid-PIN rejection check.

Appendix A. UVM log

```
UVM_INFO alarm_output_passive_monitor.sv(71) @ 5: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 7: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 7: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 15: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 21: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 21: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 25: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 35: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 35: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 35: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 45: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 49: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 49: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO keypad_checker.sv(188) @ 50: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] Waiting for pin_bad response for attempt 1/3
UVM_INFO alarm_output_passive_monitor.sv(71) @ 55: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 63: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 63: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 65: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 75: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_checker.sv(214) @ 75: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] Actual DUT pin_bad=1 for attempt 1
UVM_INFO keypad_checker.sv(221) @ 75: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] PASS: DUT rejected invalid PIN attempt 1/3
UVM_INFO keypad_checker.sv(188) @ 75: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] Waiting for pin_bad response for attempt 2/3
UVM_INFO sensor_monitor.sv(42) @ 77: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 77: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 85: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_monitor.sv(48) @ 85: uvm_test_top.alarm_env.keypad_agent.keypad_monitor [KEYPAD_MON] Observed pin_valid=1 pin_code=edcb
UVM_INFO keypad_checker.sv(95) @ 85: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] Invalid PIN stimulus observed: PIN=edcb
UVM_INFO alarm_reference_model.sv(336) @ 85: uvm_test_top.alarm_env.ref_model [REF_MODEL] KEYPAD: INVALID PIN expected, fail_count=1, locked=0
UVM_INFO pin_invalid_sequence.sv(40) @ 85: uvm_test_top.alarm_env.keypad_agent.keypad_sequencer@@seq [PIN_INVALID_SEQ] Invalid attempt 1/3: PIN=edcb
UVM_INFO sensor_monitor.sv(42) @ 91: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 91: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 95: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 105: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 105: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 105: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 115: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_checker.sv(214) @ 115: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] Actual DUT pin_bad=1 for attempt 2
UVM_INFO keypad_checker.sv(221) @ 115: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] PASS: DUT rejected invalid PIN attempt 2/3
UVM_INFO keypad_checker.sv(188) @ 115: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] Waiting for pin_bad response for attempt 3/3
UVM_INFO sensor_monitor.sv(42) @ 119: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 119: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 125: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_monitor.sv(48) @ 125: uvm_test_top.alarm_env.keypad_agent.keypad_monitor [KEYPAD_MON] Observed pin_valid=1 pin_code=a5a5
UVM_INFO keypad_checker.sv(95) @ 125: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] Invalid PIN stimulus observed: PIN=a5a5
UVM_INFO alarm_reference_model.sv(336) @ 125: uvm_test_top.alarm_env.ref_model [REF_MODEL] KEYPAD: INVALID PIN expected, fail_count=2, locked=0
UVM_INFO pin_invalid_sequence.sv(40) @ 125: uvm_test_top.alarm_env.keypad_agent.keypad_sequencer@@seq [PIN_INVALID_SEQ] Invalid attempt 2/3: PIN=a5a5
UVM_INFO sensor_monitor.sv(42) @ 133: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 133: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 135: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
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UVM_INFO alarm_output_passive_monitor.sv(71) @ 145: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 147: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 147: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 155: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_checker.sv(214) @ 155: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] Actual DUT pin_bad=1 for attempt 3
UVM_INFO keypad_checker.sv(221) @ 155: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] PASS: DUT rejected invalid PIN attempt 3/3
UVM_INFO sensor_monitor.sv(42) @ 161: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 161: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 165: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_monitor.sv(48) @ 165: uvm_test_top.alarm_env.keypad_agent.keypad_monitor [KEYPAD_MON] Observed pin_valid=1 pin_code=5a5a
UVM_INFO keypad_checker.sv(95) @ 165: uvm_test_top.alarm_env.keypad_chk [KEYPAD_INVALID] Invalid PIN stimulus observed: PIN=5a5a
UVM_INFO alarm_reference_model.sv(320) @ 165: uvm_test_top.alarm_env.ref_model [REF_MODEL] KEYPAD: INVALID PIN expected, fail_count=3, LOCKOUT expected=1
UVM_INFO pin_invalid_sequence.sv(40) @ 165: uvm_test_top.alarm_env.keypad_agent.keypad_sequencer@@seq [PIN_INVALID_SEQ] Invalid attempt 3/3: PIN=5a5a
UVM_INFO alarm_output_passive_monitor.sv(71) @ 175: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 175: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 175: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 185: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 189: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 189: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 195: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 203: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 203: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 205: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 215: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 217: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 217: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 225: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 231: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 231: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 235: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 245: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 245: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 245: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 255: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 259: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 259: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 265: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 273: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 273: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 275: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_objection.svh(1276) @ 285: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_report_server.svh(904) @ 285: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

SMART HOME ALARM CONTROLLER
KEYPAD LOCKOUT RTL BUG REPORT

UVM Verification Report — Complete Simulation Log Included

1. Objective

Verify that the keypad enters lockout after three consecutive invalid PIN attempts. The UVM test checks the expected lockout state against the DUT's STATUS register without modifying RTL.

2. Test Configuration

- UVM test: pin_lockout_test
- Stimulus sequence: pin_lockout_sequence
- Invalid PIN applied: 16'h9999 (three attempts)
- STATUS register address: 0x00000004
- Lockout status field: STATUS[6] (1 = locked, 0 = unlocked)

3. Test Procedure

1. Release system reset.
2. Drive three invalid PIN attempts through the keypad sequence.
3. Observe pin_valid, pin_code, pin_bad, and lockout_start in the waveform.
4. Read the STATUS register through AHB after the sequence.
5. Compare the DUT status against the reference model's expected lockout state.

4. Waveform Observations

The supplied waveform shows three pin_valid pulses. pin_code is 16'h9999 during each attempt and returns to 16'h0000 between attempts. pin_bad pulses alongside the invalid attempts. The displayed lockout_start signal remains low (0) during the shown interval.

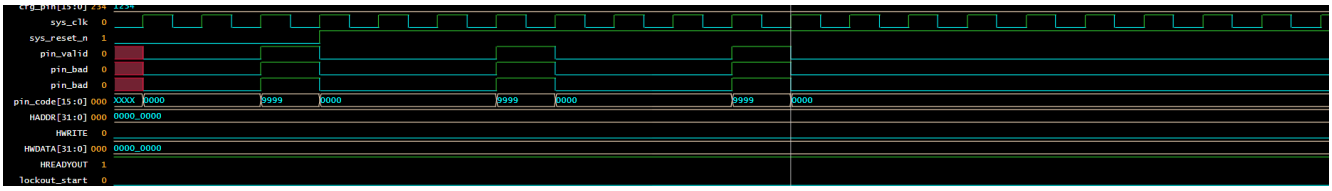


Figure 1. Waveform captured during the keypad lockout test.

5. Key Log Evidence and Bug Identification

After the third invalid PIN, the reference model reports:

```
KEYPAD: INVALID PIN expected, fail_count=3, LOCKOUT expected=1
```

The AHB STATUS read returns RDATA=0x00000100. The checker then reports:

Expected locked=1 | Actual STATUS[6]=0

FAIL: Expected lockout=1 | Actual STATUS[6]=0

BUG IDENTIFIED: The reference model expects the keypad to be locked, but the DUT reports STATUS[6]=0 (unlocked). This is a detected mismatch and is evidence of a likely RTL keypad-lockout defect.

6. Expected vs. Actual Results

Check	Expected	Observed
Invalid PIN attempts	Three invalid attempts	Three pin_valid pulses with PIN 9999
Reference model	Lockout expected after attempt 3	fail_count=3; LOCKOUT expected=1
lockout_start	Assert if this is the specified lockout activation signal	Remains low in supplied waveform
STATUS[6]	1 (locked)	0 (unlocked); STATUS read = 0x100
UVM checker	**PASS when lockout is observed**	**BUG IDENTIFIED — expected locked, actual unlocked**

7. Conclusion

The pin_lockout_test applies three invalid PIN attempts. The reference model expects lockout, but the DUT's STATUS register read returns 0x00000100, making STATUS[6]=0. The keypad checker flags the mismatch. The waveform also shows lockout_start remaining low during the displayed interval. Together, the log and waveform support reporting a keypad lockout RTL bug. Confirm that lockout_start is the intended DUT lockout activation signal before using it as standalone proof.

No RTL changes are proposed. Keep the test and checker active so this defect remains detectable.

Appendix A — Complete Simulation Log

The complete log supplied for this run is reproduced below. Bug-related UVM_ERROR entries are colored red for visibility; the log message text is retained.

```
UVM_INFO pin_lockout_test.sv(21) @ 0: uvm_test_top [PIN_LOCKOUT_TEST] Starting PIN lockout test
UVM_INFO alarm_output_passive_monitor.sv(71) @ 5: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 7: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 7: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 15: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 21: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 21: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 25: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 35: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_monitor.sv(48) @ 35: uvm_test_top.alarm_env.keypad_agent.keypad_monitor [KEYPAD_MON] Observed pin_valid=1 pin_code=9999
UVM_INFO alarm_reference_model.sv(336) @ 35: uvm_test_top.alarm_env.ref_model [REF_MODEL] KEYPAD: INVALID PIN expected, fail_count=1,
locked=0
UVM_INFO sensor_monitor.sv(42) @ 35: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 35: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 45: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 49: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 49: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 55: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 63: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 63: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 65: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 75: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_monitor.sv(48) @ 75: uvm_test_top.alarm_env.keypad_agent.keypad_monitor [KEYPAD_MON] Observed pin_valid=1 pin_code=9999
UVM_INFO alarm_reference_model.sv(336) @ 75: uvm_test_top.alarm_env.ref_model [REF_MODEL] KEYPAD: INVALID PIN expected, fail_count=2,
locked=0
UVM_INFO sensor_monitor.sv(42) @ 77: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 77: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 85: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 91: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 91: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 95: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 105: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 105: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 105: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 115: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO keypad_monitor.sv(48) @ 115: uvm_test_top.alarm_env.keypad_agent.keypad_monitor [KEYPAD_MON] Observed pin_valid=1 pin_code=9999
UVM_INFO alarm_reference_model.sv(320) @ 115: uvm_test_top.alarm_env.ref_model [REF_MODEL] KEYPAD: INVALID PIN expected,
fail_count=3, LOCKOUT expected=1
UVM_INFO sensor_monitor.sv(42) @ 119: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 119: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 125: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 133: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 133: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 135: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO pin_lockout_sequence.sv(74) @ 135: uvm_test_top.alarm_env.keypad_agent.keypad_sequencer@@pin_seq [PIN_LOCKOUT_SEQ] 3 INVALID
PIN attempts sent
UVM_INFO alarm_output_passive_monitor.sv(71) @ 145: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 147: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
```

UVM_INFO alarm_reference_model.sv(231) @ 147: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 155: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 161: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 161: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 165: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 175: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 175: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 175: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 185: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 189: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 189: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 195: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 203: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 203: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 205: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 215: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 217: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 217: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 225: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 231: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 231: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 235: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 245: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 245: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 245: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 246: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000004 WRITE=0 HTRANS=10
UVM_INFO ahb_monitor.sv(104) @ 252: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000004 WDATA=00000000 RDATA=00000100 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 252: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000004 WRITE=0 RDATA=00000100
UVM_INFO ahb_checker.sv(172) @ 252: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] STATUS READ: ACTUAL=00000100; status validation delegated to dedicated checker
UVM_INFO ahb_checker.sv(467) @ 252: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000004 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO keypad_checker.sv(172) @ 252: uvm_test_top.alarm_env.keypad_chk [KEYPAD_LOCKOUT] Expected locked=1 | Actual STATUS[6]=0
UVM_ERROR keypad_checker.sv(192) @ 252: uvm_test_top.alarm_env.keypad_chk [KEYPAD_LOCKOUT] FAIL: Expected lockout=1 | Actual STATUS[6]=0
UVM_INFO alarm_reference_model.sv(565) @ 252: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=4 RDATA=100
UVM_INFO scoreboard.sv(27) @ 252: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000004 DATA=00000000 RDATA=00000100 RESP=0
UVM_INFO sensor_checker.sv(157) @ 252: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000004 | WRITE=0 | RDATA=00000100 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 255: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 258: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000004 WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 259: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 259: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(104) @ 264: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000004 WDATA=00000000 RDATA=00000100 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 264: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000004 WRITE=0 RDATA=00000100
UVM_INFO ahb_checker.sv(172) @ 264: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] STATUS READ: ACTUAL=00000100; status validation delegated to dedicated checker

UVM_INFO ahb_checker.sv(467) @ 264: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000004 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO keypad_checker.sv(172) @ 264: uvm_test_top.alarm_env.keypad_chk [KEYPAD_LOCKOUT] Expected locked=1 | Actual STATUS[6]=0
UVM_ERROR keypad_checker.sv(192) @ 264: uvm_test_top.alarm_env.keypad_chk [KEYPAD_LOCKOUT] FAIL: Expected lockout=1 | Actual STATUS[6]=0
UVM_INFO alarm_reference_model.sv(565) @ 264: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=4 RDATA=100
UVM_INFO scoreboard.sv(27) @ 264: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000004 DATA=00000000 RDATA=00000100 RESP=0
UVM_INFO sensor_checker.sv(157) @ 264: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000004 | WRITE=0 | RDATA=00000100 | RESP=0
UVM_INFO ahb_status_sequence.sv(25) @ 264: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@status_seq [AHB_STATUS] STATUS read: addr=4 rdata=100 resp=0
UVM_INFO pin_lockout_test.sv(44) @ 264: uvm_test_top [PIN_LOCKOUT_TEST] Completed PIN lockout test
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_objection.svh(1276) @ 264: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_report_server.svh(904) @ 264: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

Appendix B — Evidence Checklist

- Waveform screenshot included.
- Complete supplied UVM simulation log included.
- Bug-related mismatch messages highlighted in red; expected lockout messages highlighted in amber.
- Recommended additional evidence: waveform view of AHB STATUS read showing HADDR=0x004, HWRITE=0, HRDATA=0x100, and ready/response signals.